



MACHINE LEARNING · ENVIRONMENTAL AI

Water Quality Classification System

Using Random Forest and Support Vector Machine

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Guide

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THE PROBLEM

Assessing water safety is slow, manual, and hard to scale



Lab-dependent testing

Traditional water-quality testing needs lab equipment, trained personnel, and considerable time.



Many parameters at once

Water quality depends on multiple physical and chemical parameters that interact with each other.



Manual analysis is slow

Checking each parameter by hand is difficult and time-consuming, especially at scale.



A gap worth closing

Machine Learning can learn patterns in existing water-quality data to predict potability automatically.

WHY THIS PROJECT

Motivation

Applying Machine Learning to a real, everyday environmental problem — while building practical ML workflow experience.



- 1 Support faster preliminary assessment of water potability
- 2 Reduce dependence on time-consuming manual analysis
- 3 Analyze multiple water-quality parameters simultaneously
- 4 Apply Machine Learning to a real-world environmental problem
- 5 Compare different classification algorithms to find an effective model
- 6 Provide a simple prediction system for preliminary assessment
- 7 Gain practical understanding of the complete ML workflow

25+ research papers studied

Approaches span classical ML to deep learning and explainability — reflecting a broad research landscape, not one shared dataset.



Main approaches found

Support Vector Machine

Random Forest

Logistic Regression

Decision Tree

K-Nearest Neighbors

Gradient Boosting

XGBoost

Artificial Neural Networks

CNN / LSTM

Ensemble Learning

Explainable AI

PCA & Feature Selection



Common limitations

- Limited dataset size
- Dataset imbalance
- Missing water-quality values
- Dependence on specific regions
- Limited generalization to other datasets
- High compute needs for deep learning
- Dependence on specific parameters

What the literature leaves open

- 1 Traditional ML models are comparatively simple and computationally efficient
- 2 Different studies report different performance depending on dataset and preprocessing
- 3 Missing values and class imbalance can affect classification performance
- 4 Many studies focus on specific geographic regions or single water sources
- 5 Deep-learning approaches are powerful but generally need more compute
- 6 Some studies use complex models without a simple prediction interface



Our focus: a consistent, standardized comparison of Random Forest and SVM for water-potability classification on a shared dataset and evaluation procedure.

DATASET

Water Potability Dataset

Source: Kaggle · Tabular water-quality dataset

3,276

Total records

10

Total columns

9

Input features

2

Target classes



Input Features

- pH
- Chloramines
- Organic Carbon
- Hardness
- Sulfate
- Trihalomethanes
- Solids
- Conductivity
- Turbidity

Target: Potability

0 → Non-Potable

1 → Potable

What the preprocessing pipeline does

1

Load the Water Potability dataset

2

Inspect structure & dimensions

3

Check missing values

4

Check duplicate records

5

Analyze data types

6

Separate features & target

7

Statistical analysis

8

Feature correlation analysis

9

Handle missing values

10

Feature scaling where required

11

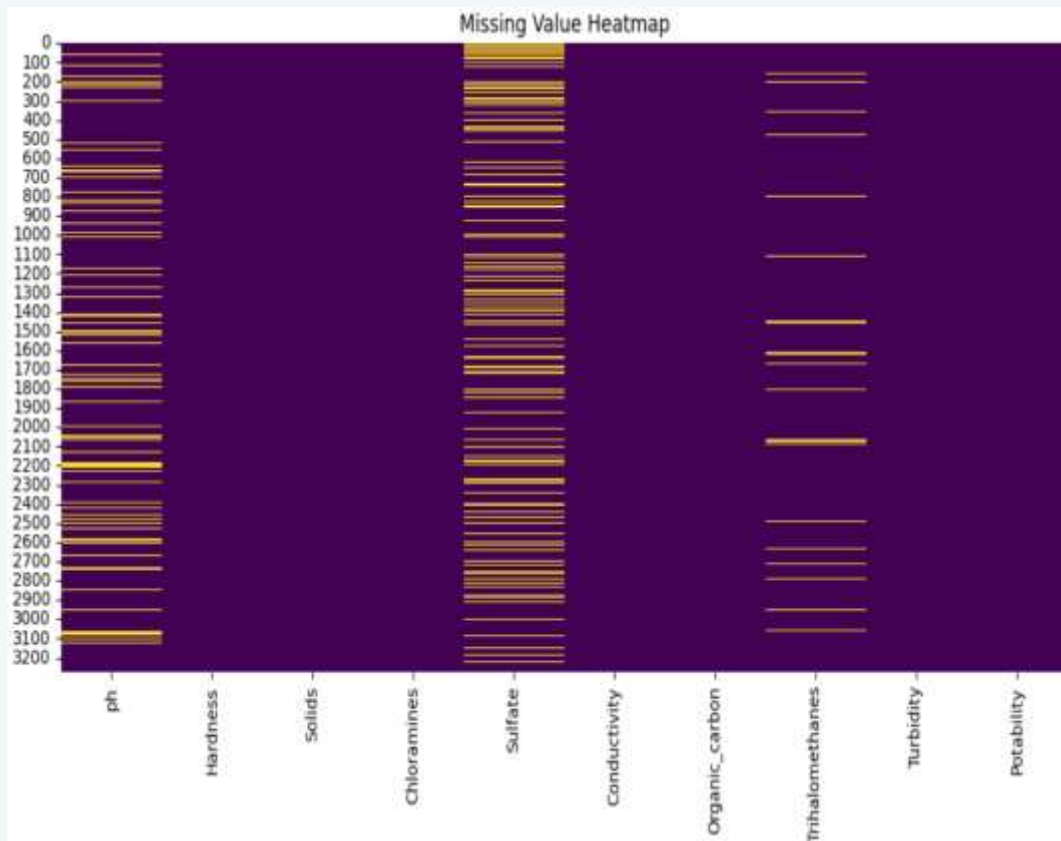
Train-test split



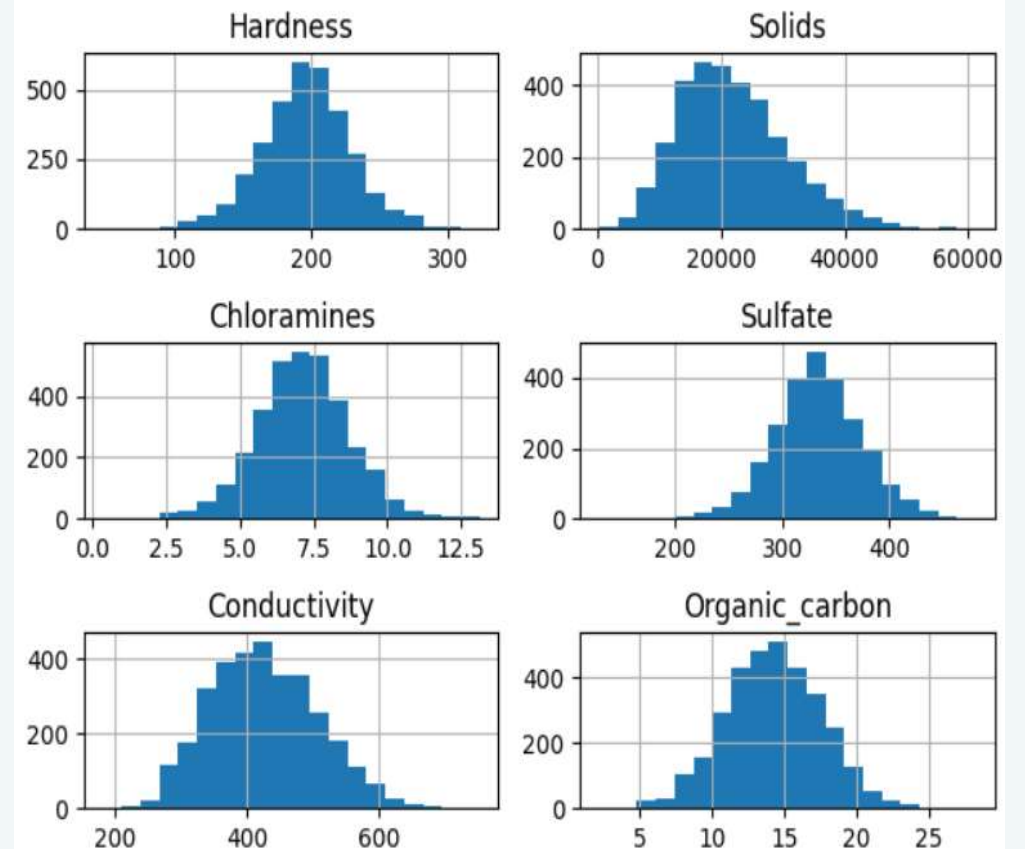
Dataset quality analysis: missing values identified & handled · duplicates checked & removed if necessary · class distribution analyzed · feature relationships examined via correlation.

The preprocessing pipeline

Missing Value Heatmap

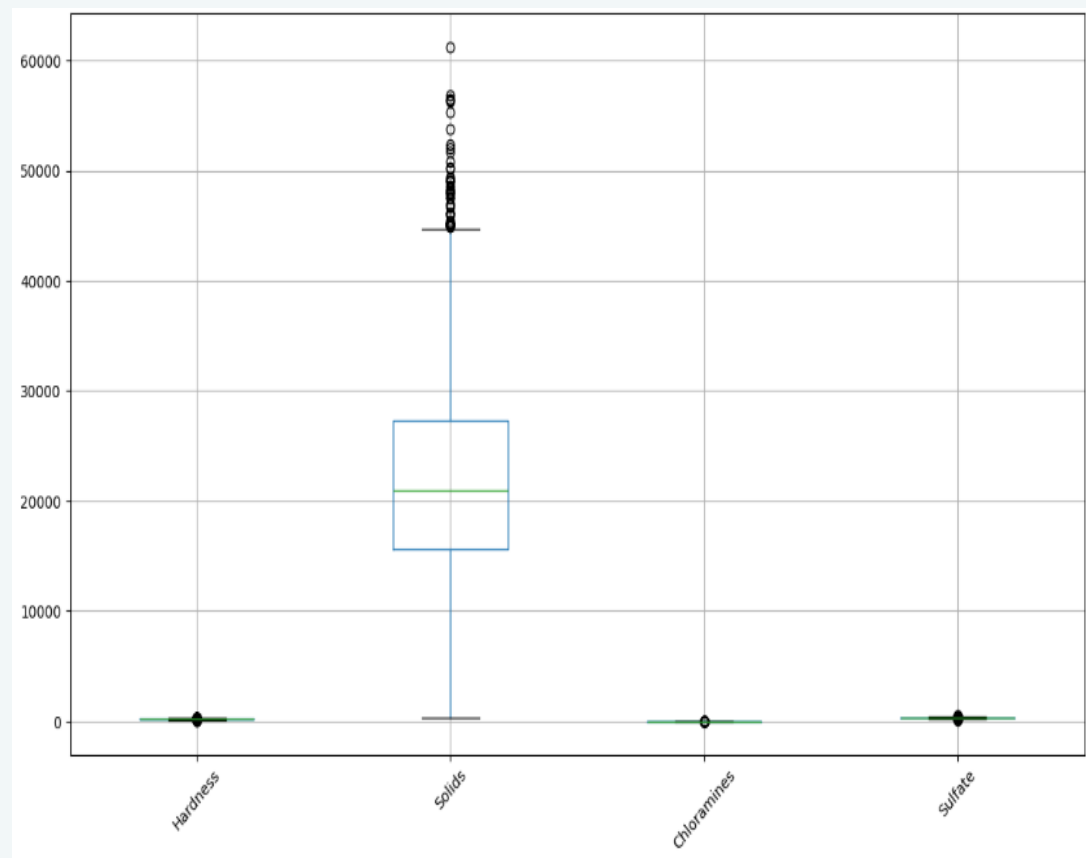


Distribution of Water Quality Features

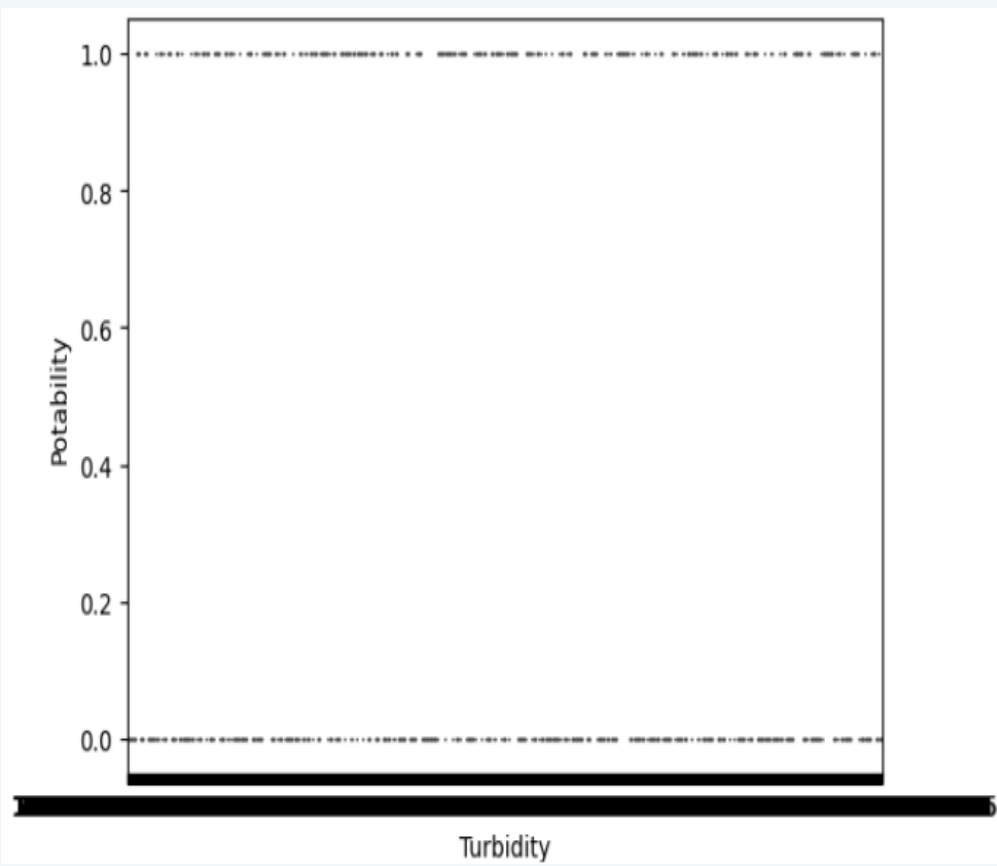


The preprocessing pipeline

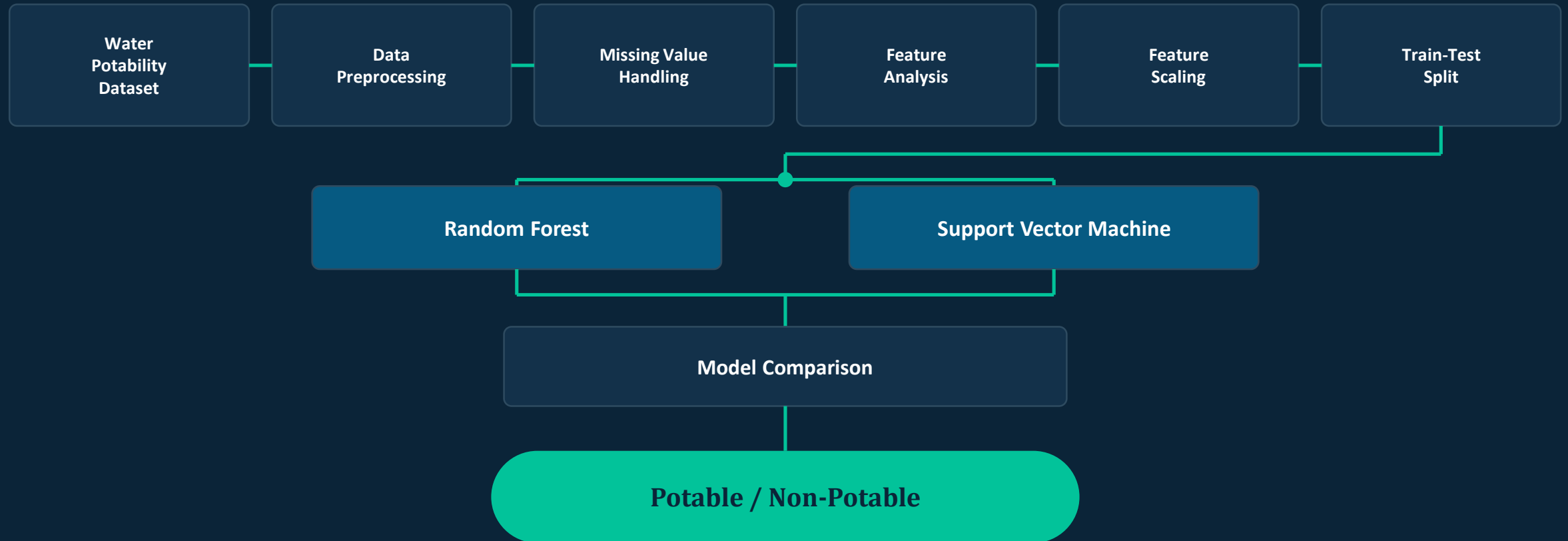
Correlation Heatmap of Water Quality Features



Boxplot Analysis of Water Quality Features



From raw dataset to a potability prediction



Final output: Water Sample → Potable or Non-Potable

Random Forest vs. Support Vector Machine



Random Forest

- Combines multiple decision trees into one prediction
- Captures nonlinear relationships between parameters
- Generally robust to noise and dataset variation
- Suited to multi-feature classification problems

Water Quality Features → Potability Prediction



Support Vector Machine

- Finds an optimal boundary between classes
- Effective for complex decision boundaries
- Feature scaling matters for good performance
- Classifies samples as potable or non-potable

Water Quality Features → Potability Prediction

Evaluated with: Accuracy · Precision · Recall · F1-Score · Confusion Matrix · ROC-AUC

Where this system can go next



Larger, diverse datasets

Use larger and more diverse water-quality datasets.



Real-time sensor data

Incorporate real-time water-quality sensor readings.



Web / mobile app

Develop a web or mobile application for easy prediction.



IoT integration

Integrate IoT sensors for continuous water monitoring.



More parameters

Include additional water-quality parameters.



Advanced models

Explore advanced Machine Learning and Deep Learning models.



Explainable AI

Add XAI to explain why a sample is potable or not.



Cross-region coverage

Extend monitoring across different geographic regions.

CONCLUSION

A practical step toward smarter water assessment

- Water quality assessment is a key environmental & public-health problem
- ML can classify water samples from their physical and chemical characteristics
- The system uses the Water Potability Dataset for classification
- Random Forest and SVM were implemented and evaluated using multiple classification metrics
- Performance will be evaluated using multiple classification metrics
- The best-performing model can power a user-friendly prediction system

The system supports preliminary decision-making — laboratory testing remains necessary for definitive water-safety verification.



Thank You